

COMPUTER NETWORKS

Week-12

Secure Communication

- There are certain desirable properties of secure communication:
- Confidentiality
- Message Integrity
- End-Point Authentication
- Operational Security

Confidentiality

- Only the sender and intended receiver should be able to understand the contents of the transmitted message
- This requires that the message is **encrypted** so that an intercepted message cannot be understood by an interceptor
- → secure communication

Message integrity

- The content of communication is not altered, either maliciously or by accident, **in transit**
- Extensions to the check summing techniques that we encountered in reliable transport and data link protocols can be used to provide such message integrity

End-point authentication

- Both the sender and receiver should be able to confirm the identity of the other party involved in the communication
- When a user wants to access an inbox, how does the mail server verify that the user is the person he or she claims to be?

Operational security

- Almost all organizations (companies, universities, etc.) today have networks that are attached to the public Internet
- Attackers can attempt to deposit worms into the hosts in the network, obtain corporate secrets, map the internal network configurations, and launch DoS (denial of Service) attacks
- Operational devices such as firewalls and intrusion detection systems are used to counter attacks against an organization's network

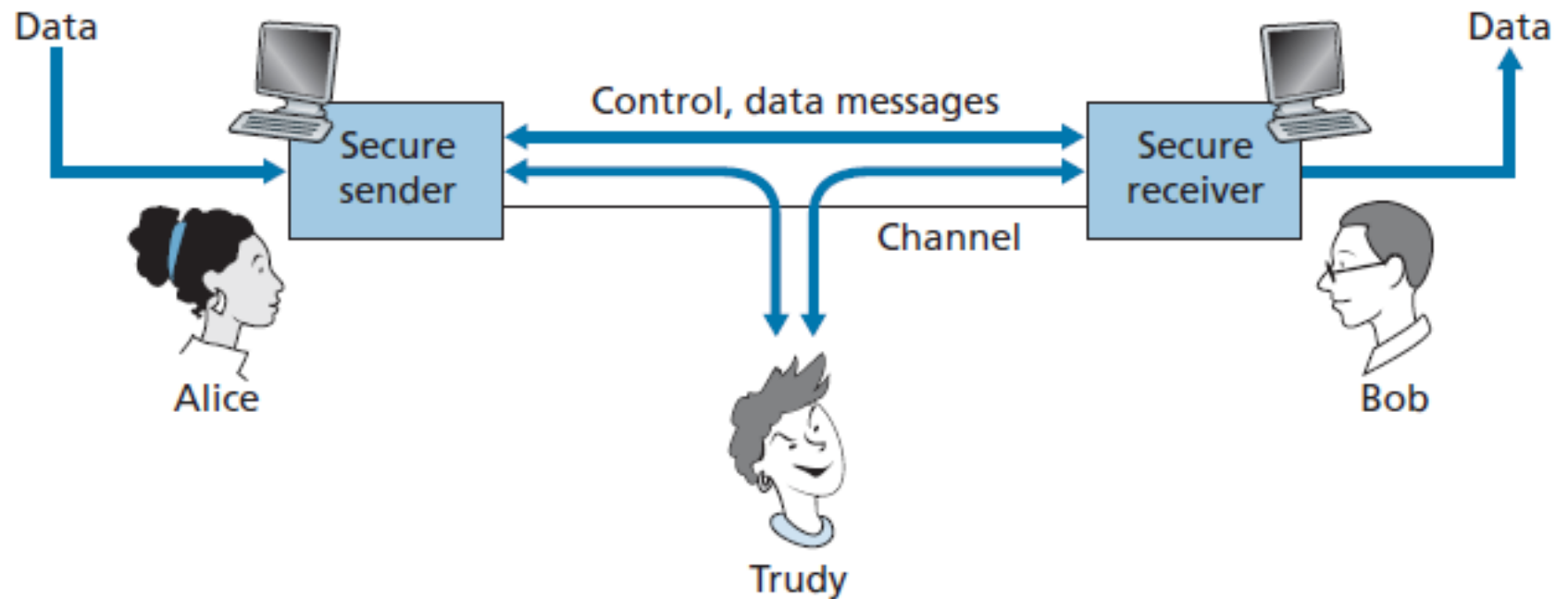
Operational security

- A firewall sits between the organization's network and the public network, controlling packet access to and from the network
- An intrusion detection system is a monitoring system that detects suspicious activities and generates alerts for the network administrators about suspicious activity

Cryptography

- Cryptographic techniques allow a sender to disguise data so that an intruder can gain no information from the intercepted data
- The receiver, of course, must be able to recover the original data from the disguised data

Intruder



Intruder

- Alice and Bob will exchange control messages and data messages to ensure:
- Secure data exchange
- Confidentiality
- End-point authentication
- Message integrity
- Examples
- A network user might want to transfer his credit card number securely to a Web server to purchase an item online
- Or a user want to interact with his bank online

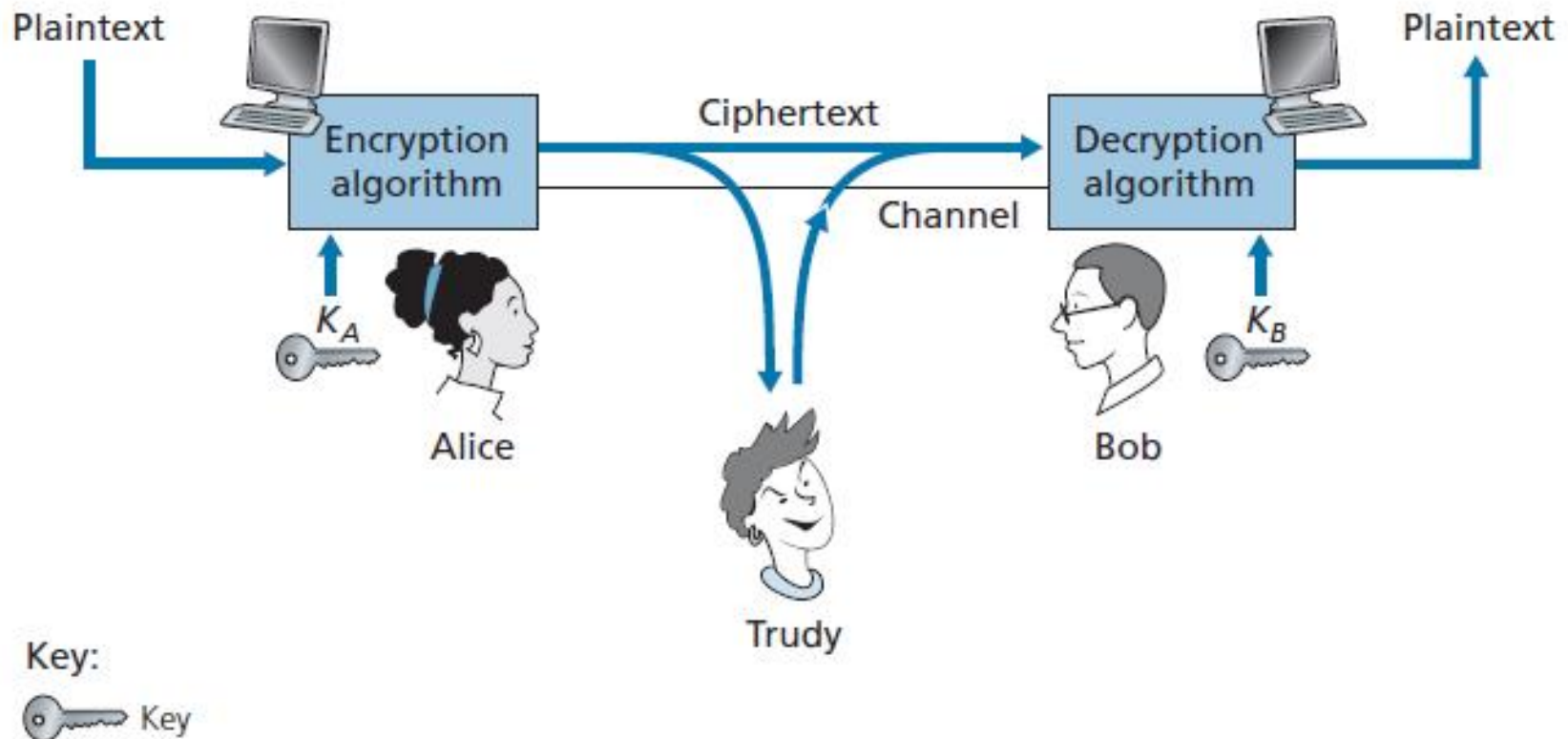
Intruder

- An intruder can potentially perform:
- *eavesdropping*—sniffing and recording control and data messages on the channel
- *modification, insertion, or deletion* of messages or message content

Cryptography

- Cryptography has a long history dating back at least as far as Julius Caesar
- Modern cryptographic techniques, including many of those used in the Internet, are based on advances made in the past 30 years
- Cryptographic techniques allow a sender to disguise data so that an intruder can gain no information from the intercepted data
- The receiver, of course, must be able to recover the original data from the disguised data

Components of Cryptography



Cryptography

- Alice's message in its original form is known as **plaintext**, or **cleartext**
- Alice encrypts her plaintext message using an **encryption algorithm** so that the encrypted message, known as **ciphertext**, has no meaning for any intruder

Cryptography

- The encryption technique itself is *known*—published, standardized, and available to everyone, even a potential intruder!
- If everyone knows the method for encoding data, then there must be some secret information that prevents an intruder from decrypting the transmitted data
- → a key is used

Encryption

- Sender provides a **key**, a string of numbers or characters, as input to the encryption algorithm
- The encryption algorithm takes the key and the plaintext message, m , as input and produces **ciphertext** as output
- The notation
- $K_A(m)$ refers to the ciphertext form (encrypted using the key K_A) of the plaintext message, m

Encryption

- Bob will provide a key, K_B , to the decryption algorithm
- The decryption algorithm takes the ciphertext and Bob's key as input and produces the original plaintext as output
- That is, if Bob receives an encrypted message $K_A(m)$, he decrypts it by computing:
- $K_B(K_A(m)) = m$

Encryption & Decryption

- In **symmetric key systems**, sender and receiver's keys are identical and are secret
- In **public key systems**, a pair of keys is used
- One of the keys is known to both Sender and the Receiver (indeed, it is known to the whole world)
- The other key is known only by either Receiver or Sender (but not both)

Symmetric Key Systems

- Encrypted communication requires that the two communicating parties share a **common key** for encryption and decryption
- The two parties must somehow agree on the shared key → through a **secure** communication **channel**
- In a networked world, however, communicating parties may never meet and may never converse except over the network → need for encryption

Symmetric Key Cryptography

- A very old, very simple symmetric key algorithm attributed to Julius Caesar, known as the **Caesar cipher**
- This is a cipher is a method for encrypting data
- For English text, the Caesar cipher would work by taking each letter in the plaintext message and substituting the letter that is k letters later (allowing wraparound; that is, having the letter z followed by the letter a) in the alphabet

Symmetric Key Cryptography

- Caesar Cipher
- Monoalphabetic Cipher

Caesar cipher

- If $k = 3$, then the letter a in plaintext becomes d in ciphertext
- Letter b in plaintext becomes e in ciphertext, and so on
- Here, the value of k serves as the key
- Let's convert the following into cipher using the above key:

Your network is your net worth

Symmetric Key Cryptography

- An improvement on the Caesar cipher is the **monoalphabetic cipher**
- It substitutes one letter of the alphabet with another letter of the alphabet
- There is no need to have a regular pattern (e.g., substitution with an offset of k for all letters)
- Any letter can be substituted for any other letter, as long as each letter has a unique substitute letter, and vice versa

Monoalphabetic Cipher

Plaintext letter: a b c d e f g h i j k l m n o p q r s t u v w x y z
Ciphertext letter: m n b v c x z a s d f g h j k l p o i u y t r e w q

- Better than the Caesar cipher
- As there are $26!$ (on the order of 10^{26}) possible pairings of letters rather than 25 possible pairings

Monoalphabetic Cipher

Plaintext letter: a b c d e f g h i j k l m n o p q r s t u v w x y z
Ciphertext letter: m n b v c x z a s d f g h j k l p o i u y t r e w q

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Monoalphabetic Cipher

- A monoalphabetic cipher is not better but it is definitely more secure and flexible than a Caesar cipher

Caesar Cipher

- Only 25 possible shifts (since shifting by 26 gives original text)
- Easier to understand and implement
- Simple shifting rule (A→D, B→E, etc.)
- Pattern is predictable → Easily broken by brute force

Monoalphabetic Cipher

- Uses a full permutation of the alphabet
- Total keys = $26!$ ($\approx 4 \times 10^{26}$ possibilities)
- Practically impossible to brute-force manually
- More complex key management
- Each letter maps randomly (e.g., $A \rightarrow Q$, $B \rightarrow M$, $C \rightarrow Z \dots$)
- No fixed pattern $\rightarrow$ harder to guess $\rightarrow$ Requires deeper analysis

Ex-1

- Plaintext: HELLO
- Caesar (shift +3): KHOOR
- Monoalphabetic (random mapping): XQZZY
- Notice how monoalphabetic output looks much less predictable

Ex-2

- The given ciphertext is PHHW PH DIWHU FODVV
It is known to be encoded using a Caesar cipher
- Find the plaintext

Ex-2

- The given ciphertext is PHHW PH DIWHU FODVV
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- Find the plaintext
- Try common shifts

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- Try shift -3

Ex-2

- The given ciphertext is **PHHW PH DIWHU FODVV**
It is known to be encoded using a Caesar cipher
- Find the plaintext
- Try common shifts
- Try shift -3:
- $P \rightarrow M$
- $H \rightarrow E$
- $H \rightarrow E$
- $W \rightarrow T$

Ex-2

- The given ciphertext is **PHHW PH DIWHU FODVV**
It is known to be encoded using a Caesar cipher
- Find the plaintext
- Try common shifts
- Try shift -3:
- $P \rightarrow M$
- $H \rightarrow E$
- $H \rightarrow E$
- $W \rightarrow T$
- **MEET ME AFTER CLASS**

Ex-3

- A school uses a Caesar cipher with a shift of +2 to send short messages
- Message sent: QPNKPG ENCUU PQY
- Decrypt the message
- Explain one limitation of using Caesar cipher

Ex-3

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- Explain one limitation of using Caesar cipher

[illegible]

Ex-4

- A student claims to have encrypted the message: SUCCESS using a monoalphabetic cipher and obtained: **XABBCYX**
- Verify whether the given ciphertext could have been produced using a valid monoalphabetic cipher
- Justify your answer

Ex-4

- A student claims to have encrypted the message: SUCCESS using a monoalphabetic cipher and obtained: XABBCYX
- Verify whether the given ciphertext could have been produced using a valid monoalphabetic cipher. Justify your answer.
- Answer:
- The ciphertext is NOT valid for a monoalphabetic cipher

Ex-4

- A student claims to have encrypted the message: SUCCESS using a monoalphabetic cipher and obtained: XABBCYX
- Verify whether the given ciphertext could have been produced using a valid monoalphabetic cipher. Justify your answer.
- The ciphertext is NOT valid for a monoalphabetic cipher
- Check consistency:
- Plaintext: S U C C E S S
Ciphertext: X A B B C Y X

Ex-4

- A student claims to have encrypted the message: SUCCESS using a monoalphabetic cipher and obtained: XABBCYX
- Verify whether the given ciphertext could have been produced using a valid monoalphabetic cipher. Justify your answer.
- The ciphertext is NOT valid for a monoalphabetic cipher
- Check consistency:
- Plaintext: S U C C E S S
Ciphertext: X A B B C Y X
- Now track mappings:
- $S \rightarrow X$
- $S \rightarrow Y$ (inconsistent)

Ex-4

- Plaintext: S U C C E S S
Ciphertext: X A B B C Y X
- Now track mappings:
- $S \rightarrow X$
- $S \rightarrow Y$ (**inconsistent!**)
- Same plaintext letter maps to different ciphertext letters $\rightarrow$ **violation**
- Not a valid monoalphabetic cipher
- **In monoalphabetic cipher: Each letter must map to one fixed letter only**

Vigenère Cipher

- Introduces the idea of a key (word) instead of a single shift
 - An improved version of Caesar Cipher
 - Uses a keyword to apply different shifts to different letters
-
- Caesar = same shift every time
Vigenère = different shift per letter

Ex-5

- Use letter positions:
- A=0
- B=1
-
- Z=25

Ex-5

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Ex-5

- Plain Text

HELLO WORLD

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Ex-5

- Plain Text

HELLO WORLD

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Key to use	K	E	Y
	10	4	24

Ex-5

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Key to use	K	E	Y
	10	4	24

Plain Text			H	E	L	L	O		W	O	R	L	D
Letter value			7	4	11	11	14		22	14	17	11	3
Apply Key			K	E	Y	K	E		Y	K	E	Y	K

Ex-5 : Applying Key

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Plain Text			H	E	L	L	O		W	O	R	L	D
Letter value			7	4	11	11	14		22	14	17	11	3
Apply Key			K	E	Y	K	E		Y	K	E	Y	K
			10	4	24	10	4		24	10	4	24	10
Add			17	8	35	21	18		46	24	21	35	13

Ex-5

Plain Text			H	E	L	L	O		W	O	R	L	D
Letter value			7	4	11	11	14		22	14	17	11	3
Apply Key			K	E	Y	K	E		Y	K	E	Y	K
			10	4	24	10	4		24	10	4	24	10
Add			17	8	35	21	18		46	24	21	35	13

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Ex-5 : Wrap around : use mod 26

Plain Text	H	E	L	L	O	W	O	R	L	D
Letter value	7	4	11	11	14	22	14	17	11	3
Apply Key	K	E	Y	K	E	Y	K	E	Y	K
	10	4	24	10	4	24	10	4	24	10
Add	17	8	35	21	18	46	24	21	35	13
Wrap around mod 26	17	8	9	21	18	20	24	21	9	13

Ex-5

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Plain Text			H	E	L	L	O		W	O	R	L	D
Letter value			7	4	11	11	14		22	14	17	11	3
Apply Key			K	E	Y	K	E		Y	K	E	Y	K
			10	4	24	10	4		24	10	4	24	10
Add			17	8	35	21	18		46	24	21	35	13
Wrap around			17	8	9	21	18		20	24	21	9	13
mod 26													
Cipher			R	I	J	V	S		U	Y	V	J	N

Ex-5

Plain Text	H	E	L	L	O		W	O	R	L	D
Cipher	R	I	J	V	S		U	Y	V	J	N

Decrypt	Plain Text = (Cipher – Key) mod 26								

Vigenère Cipher

- In real computer networks, the Vigenère Cipher is not used anymore as a practical security mechanism in modern networks
 - It is mainly a historical and educational cipher, but it is still very useful for understanding how modern encryption evolved
 - It is vulnerable to frequency analysis
 - It is too slow for digital systems
 - It can be broken with statistical methods
 - It does not support large-scale secure communication
-
- So in real systems like:
 - banking networks
 - web traffic (HTTPS)
 - email security
- it is **not used at all**

AES (Modern Symmetric Encryption)

- Advanced Encryption Standard
- The most widely used encryption in modern networks
- Symmetric:
- AES = a locked box with one shared key
Sender locks → receiver unlocks with **same** key

AES

- Used in modern networks for sending large amount of data
- HTTPS web traffic (after connection is established)
- WhatsApp
- Banking transactions
- Cloud storage encryption
- Wi-Fi security

AES

- Very fast
- Efficient for large data
- Strong security when key is protected

RSA

- RSA stands for **Rivest-Shamir-Adleman**, named after its creators Ron Rivest, Adi Shamir, and Leonard Adleman, who first described the algorithm in 1977
- Public key cryptography
- Asymmetric Encryption
- Foundation of secure communication on internet
- Uses a public key (shared with everyone) for **encryption**
- Used a separate private key (kept secret) for **decryption**

RSA

- Used in :
 - HTTPS handshake (secure websites)
 - Digital certificates
 - Software authentication
 - Secure email systems
- Public key → anyone can use it to encrypt
- Private key → only owner can decrypt
- Private key is mathematically linked but impossible to derive from the public key in any practical time

RSA

- Public key = Locked mailbox slot → Anyone can drop a message inside (encrypt)
- Private key = Only owner can open mailbox and read messages (decrypt)
- The lock is designed so that seeing the slot does NOT reveal how to open it
- Knowing the public key does NOT help you compute the private key due to computational hardness

Hybrid Encryption : Real world Model

- **Example: HTTPS (online banking / shopping)**
 - Browser connects to website
 - RSA is used to securely exchange a **session key**
 - AES is then used for all actual data encryption
 - This is called **hybrid** encryption
-
- Modern internet security does not rely on one algorithm
 - It combines RSA for secure key exchange and AES for fast data encryption

Cyclic Redundancy Check

- One of the most popular error-checking schemes
- With CRC, a message is treated as one long binary number, which is divided by a preset number/generator
- The remainder is used as the CRC code
- a strong error-detecting code used in network protocols (like Ethernet) to detect data corruption during transmission

CRC

- The receiving hardware divides the received message by the same preset number, which generates a remainder
- The receiving hardware checks through CRC
- If remainder is **NOT** zero → the message is assumed to be in error
- In practice, the CRC algorithm is implemented using binary logic on a bit-by-bit basis to simplify memory requirements.

Logical XOR

- Instead of subtraction, XOR is used because:
- No borrowing
- Fast in hardware
- Easy to implement
- $\text{XOR} = \text{subtraction without carry/borrow}$

Logical XOR

XOR operation		
A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

Same bits --> 0

Different bits --> 1

Called difference detector

Logical XOR in Real Applications

- CRC (error detection)
- Encryption (simple ciphers)
- Parity checking
- Digital circuits

XOR - Ex1

- $1011 \oplus 1100 = ?$

XOR - Ex1

- $1011 \oplus 1100 = ?$
- 0111

XOR - Ex2

- $1111 \oplus 0101 = ?$

XOR - Ex2

- $1111 \oplus 0101 = ?$
- 1010
- Compare bits $\rightarrow$ same = 0, different = 1

Ex-3

- Message to be sent: 1101011011
- Preset generator/ divisor : 10011
- Length of generator : $L = 5$
- Bits to be appended to original message : $L - 1$
- $5 - 1 = 4$
- Data becomes : 11010110110000

Ex-3 (Sender Side)

Handwritten long division of 110000101 by 10011. The dividend is 110000101 and the divisor is 10011. The quotient is 11011000 and the remainder is 1110.

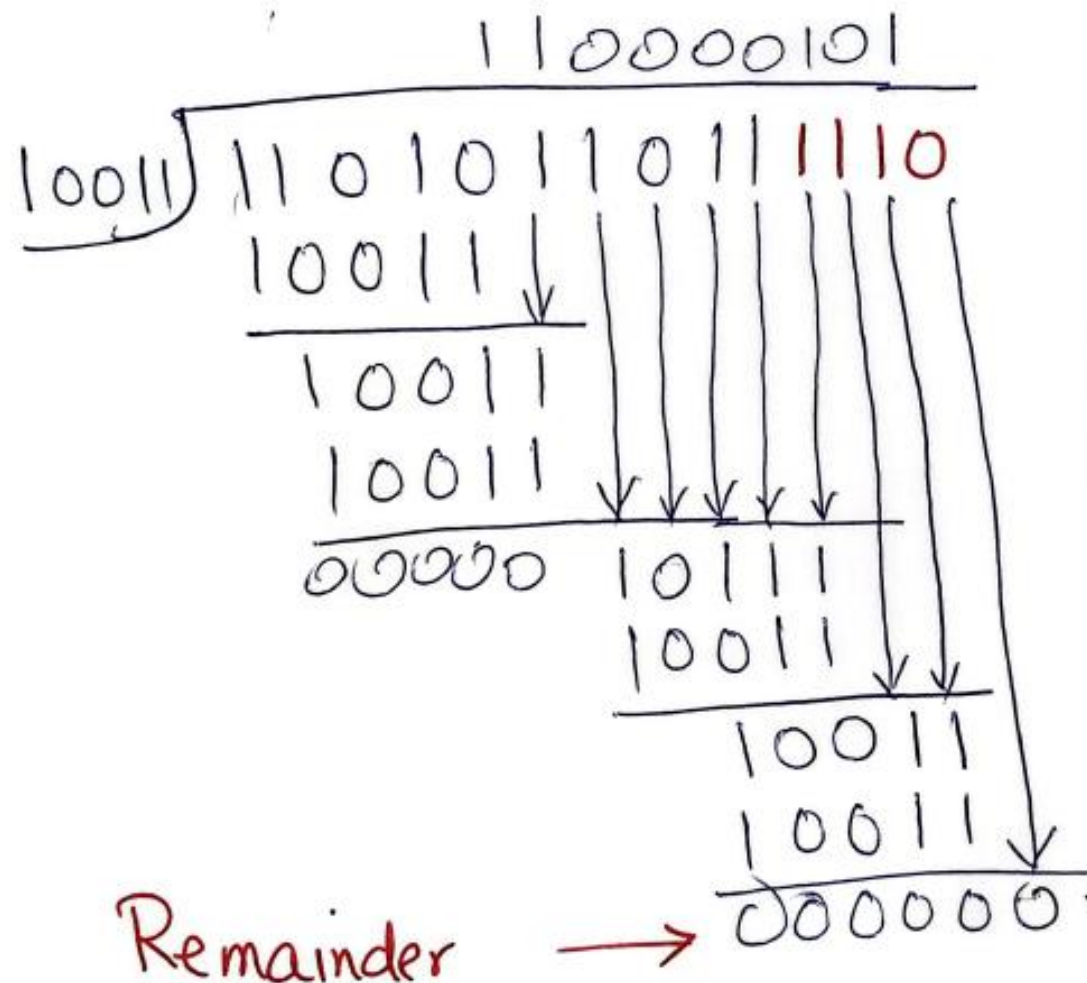
$$\begin{array}{r} 110000101 \\ \underline{10011} \\ 10011 \\ \underline{10011} \\ 00000 \\ 10110 \\ \underline{10011} \\ 10100 \\ \underline{10011} \\ 1110 \end{array}$$

Remainder $\rightarrow 1110$

Ex-3

- Append the CRC generated to the original data and carry out the division using the same divisor
- If remainder at the receiver is ZERO → No error

Ex-3 (receiver Side)



CRC

- The probability of detecting an error is 100% for all errors of the same length as the CRC or less
- For example, CRC-16 is guaranteed to detect errors if 16 or fewer bits are affected
- If the burst error is longer than the CRC, then CRC is not perfect but is close to it
- CRC-16 will detect about 99.998% of all burst errors longer than 16 bits
- CRC-32 will detect about 99.99999998% of all burst errors longer than 32 bits

Ex-4

- Data : 101100
- Generator/ Divisor : 1101
- Calculate remainder at the sender side

Ex-4

- Data : 101100
- Generator/ Divisor : 1101
- Remainder at the sender side : 111
- Append CRC to data and recalculate the remainder

Ex-4

- Data : 101100
- Generator/ Divisor : 1101
- Remainder at the sender side : 111
- At receiver side
- Remainder : ?
- Error detected OR No Error?

CRC- Summary

- CRC helps the receiver determine:
- “Did the data arrive correctly, or was it corrupted during transmission?”
- The sender:
 - Calculates a special value called the **CRC remainder**
 - Appends it to the data
 - Sends both together
- The receiver:
 - Performs the same CRC calculation
 - Checks the remainder
 - If remainder = 0 → data assumed correct
 - If remainder $\neq$ 0 → error detected

Ex-6 : To be submitted

- A father gives nothing better than good education
- By Hazrat Muhammad PBUH
- Convert it into cipher using Vigenère cipher technique
- Key : LEMON
- You can work in pairs and submit one solution

Book-Reference

- Chapter 29, Douglas E. Comer
- Chp 6 James W. Kurose